

# Complete IPv6 Guide for Network Engineers

## 1. Introduction to IPv6

IPv6 (Internet Protocol version 6) is the latest version of the Internet Protocol designed to replace IPv4. It provides a vastly larger address space, improved routing efficiency, auto-configuration, integrated security, and better support for modern networks.

## 2. Why IPv6?

IPv4 uses 32-bit addresses, allowing about 4.3 billion addresses, which are nearly exhausted. IPv6 uses 128-bit addresses, providing approximately 340 undecillion addresses. It removes the need for NAT, simplifies network design, and improves end-to-end connectivity.

## 3. IPv4 vs IPv6

IPv4: 32-bit address, dotted decimal format, NAT required, optional IPSec, broadcast supported.  
IPv6: 128-bit address, hexadecimal format, no NAT required, IPSec built-in, no broadcast (uses multicast and anycast).

## 4. IPv6 Address Format

IPv6 addresses are 128 bits long and written in hexadecimal, divided into eight 16-bit blocks separated by colons. Example: 2001:0DB8:0000:0000:0000:FF00:0042:8329 Leading zeros can be omitted, and consecutive zero blocks can be compressed using :: once per address.

## 5. Types of IPv6 Addresses

1. Unicast: Identifies a single interface. 2. Multicast: Identifies multiple interfaces. 3. Anycast: Delivered to the nearest interface. Special ranges: - Global Unicast: 2000::/3 - Link Local: FE80::/10 - Loopback: ::1 - Unique Local: FC00::/7

## 6. IPv6 Header

The IPv6 header is simplified compared to IPv4. Important fields include: - Version - Traffic Class - Flow Label - Payload Length - Next Header - Hop Limit - Source Address - Destination Address

## 7. IPv6 Address Configuration

Manual configuration: Router(config-if)# ipv6 address 2001:DB8:1::1/64 Auto configuration methods: 1. SLAAC 2. DHCPv6 3. Manual

## 8. Enabling IPv6 Routing

To enable IPv6 forwarding on Cisco routers: Router(config)# ipv6 unicast-routing

## 9. IPv6 Static Routing

Command: Router(config)# ipv6 route 2001:DB8:2::/64 2001:DB8:12::2 Used to manually configure routes to remote IPv6 networks.

## 10. IPv6 Dynamic Routing

IPv6 supports OSPFv3, EIGRP for IPv6, RIPv6, and BGP. Example OSPFv3: Router(config-if)# ipv6 ospf 10 area 0

## 11. Neighbor Discovery Protocol (NDP)

NDP replaces ARP in IPv6. It handles: - Address resolution - Router discovery - Neighbor reachability - Duplicate address detection

## 12. ICMPv6

ICMPv6 supports error reporting and NDP operations. Common messages: - Router Solicitation - Router Advertisement - Neighbor Solicitation - Neighbor Advertisement

## 13. IPv6 ACLs

Example: Router(config)# ipv6 access-list BLOCK\_PC1 Router(config-ipv6-acl)# deny ipv6 host 2001:DB8:1::10 any Router(config-ipv6-acl)# permit ipv6 any any

## 14. DHCPv6

DHCPv6 provides IPv6 addressing and additional parameters. Example: Router(config)# ipv6 dhcp pool IPV6POOL Router(config-dhcp)# address prefix 2001:DB8:1::/64

## 15. Verification Commands

show ipv6 interface brief show ipv6 route show ipv6 neighbors show ipv6 ospf neighbor show ipv6 dhcp binding

## 16. Advantages of IPv6

- Huge address space - Better routing - Auto configuration - Built-in security - No NAT - Improved multicast support

## 17. Common Interview Questions

1. Why IPv6 instead of IPv4? 2. Difference between SLAAC and DHCPv6? 3. What is link-local address? 4. How does NDP work? 5. How to configure IPv6 static route?

## **18. Real-Time IPv6 Labs**

Important practical labs: 1. Basic IPv6 addressing 2. IPv6 static routing 3. OSPFv3 configuration 4. DHCPv6 setup 5. IPv6 ACL implementation These are commonly asked in network engineer interviews.